

THE REAL POTENTIAL BEHIND MODERN TEXT TERMINALS

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About the author

Artur Twardowski

- programming since 2001
- working for Mobica (also known as Cognizant)
- C/C++, embedded, electronics, the lower, the better
  (but also Python)

Table of Contents

0. Introduction

1. Control Sequences

2. Beyond basic ASCII

3. State of the Art

Why terminals?

- easy automation of tasks
- easy prototyping and development of tools
- low bandwidth required
- low hardware requirements (both sides)

PART I: CONTROL SEQUENCES

Typewriters?



IBM 2741 (1965)



Photo by: Oscar.nierstrasz

https://en.wikipedia.org/wiki/Computer_terminal#/media/File:IBM_2741_Communications_Terminal.jpg

Typewriters?





Teletype Corporation Model 33 (1963)
DEC PDP-8 Altair 8800

Photo by: Carlo Nardone

https://upload.wikimedia.org/wikipedia/commons/2/25/DEC_PDP-8_family_-_PDP-8F_%281972%29%2C_PDP-8_%281965%29%2C_PDP-8S_%28c.1966%29_-_and_Teletype_ASR-33_%281964%29%2C_on_the_side_of_Data_General_Eclips_%281974%29_%26_Nova_%281969%29_-_Computer_History_Museum_%282007-11-10_23.08.07_by_Carlo_Nardone%29.jpg

Photo by: Tim Colegrove

https://en.wikipedia.org/wiki/Teletype_Model_33#/media/File:Altair_8800_and_Model_33_ASR_Teletype.jpg

Typewriters?

Teletype encoding standards

Baudot code (5-bit, since 1870)

ITA1, ITA2 (5-bit, since 1932)

ASCII (7-bit, since 1963)

EBCDIC (8-bit, since 1963)

ASCII standard

ASCII (ANSI_X3.4, ISO-646):

00X XXXX	0x00..0x1F	control codes
011 XXXX	0x30..0x3A	digits
10X XXXX	0x40..0x5A	uppercase letters
11X XXXX	0x60..0x7A	lowercase letters
111 1111	0x7F	DEL (RUB OUT)

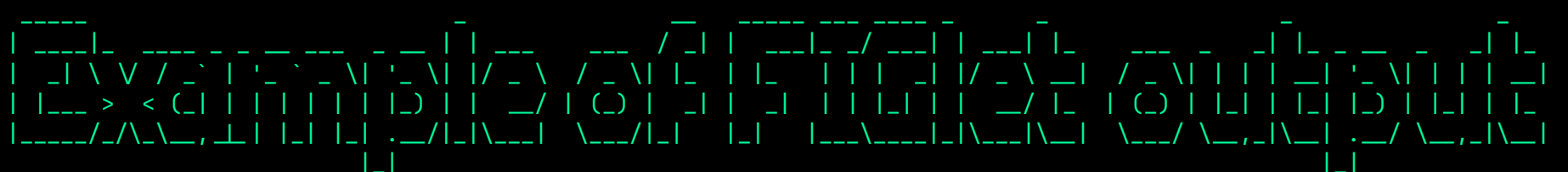
0x00	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s	s
0x20		!	"	#	$	%	&	'	(	)	*	+	,	-	.	/	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
0x40	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
0x60	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	{	|	}	~	s

Some control codes:

7	(0x07)	- bell (literally)
8	(0x08)	- backspace
10	(0x0a)	- line feed
13	(0x0d)	- carriage return
27	(0x1b)	- Escape (like '\')

ASCII Art





Typewriters?

Legacy:

TTY - Teletype / Teletypewriter

CR - Carriage Return

LF - Line Feed

Baud, baudrate

ed - editor for immortals :)

Why don't use a CRT?



Tektronix 4014 (1972)

Photo by: Piergiovanna Grossi

https://en.wikipedia.org/wiki/Tektronix_4010#/media/File:Tektronix_4010_DSC_0056.jpg

Video Display Units



Lear Siegler ADM-3A (1976)

Photo by: Daniel Sancho

https://en.wikipedia.org/wiki/Computer_terminal#/media/File:ADM_3A_(82481457).jpg



DEC VT-100 (1978)

Photo by: Autopilot (Wikipedia user)

https://en.wikipedia.org/wiki/Computer_terminal#/media/File:RT-11_help.jpg



Photo by: Santo Nucifora

https://vintagecomputer.ca/lear-siegler-adm-3a-terminal/

ANSI X3.64 (Escape Codes)

\x1b[...

Move cursor *n* characters...

ESC[*n*A - ...up
ESC[*n*B - ...down
ESC[*n*C - ...right
ESC[*n*D - ...left

Clear...

ESC[2J - entire screen
ESC[0K - current line from the cursor
ESC[2K - current line, entirely

Place cursor...

ESC[*row*;*col*H

ANSI X3.64 (Escape Codes)

\x1b[...

Select Graphic Rendition

ESC[*param1*;*param2*;*param3*;...;m

0 - reset	30..37	
1 ¦ 22 - **bold** on/off	90..97 - foreground color	
3 ¦ 23 - *italic* on/off	38	
4 ¦ 24 - underline on/off		
21 - double underline	40..47	
9 ¦ 29 - ~~strikethrough~~ on/off	100..107 - background color	
5 ¦ 25 - blink on/off	48	

ANSI X3.64 - Colors

Example:

ESC[1;31m → bright red text (bold)

30:	---	black	1;30	90:	--- bright black (!)
31:	--R	red	1;31	91:	--R bright red
32:	-G-	green	1;32	92:	-G- bright green
33:	-GR	yellow	1;33	93:	-GR bright yellow
34:	B--	blue	1;34	94:	B-- bright blue
35:	B-R	magenta	1;35	95:	B-R bright magenta
36:	BG-	cyan	1;36	96:	BG- bright cyan
37:	BGR	white (actually gray)	1;37	97:	BGR bright white (really)

ANSI X3.64 - Colors

Example:

ESC[97;41m → white text on red background

40:	--- black	100:	--- bright black
41:	--R red	101:	--R bright red
42:	-G- green	102:	-G- bright green
43:	-GR yellow	103:	-GR bright yellow
44:	B-- blue	104:	B-- bright blue
45:	B-R magenta	105:	B-R bright magenta
46:	BG- cyan	106:	BG- bright cyan
47:	BGR white	107:	BGR bright white

PART II: BEYOND BASIC ASCII

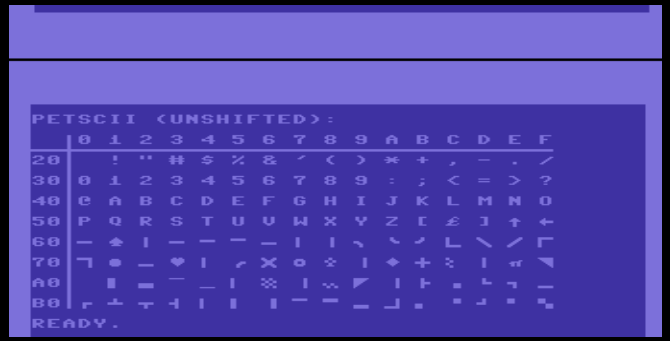
Semigraphics



CGA character set (1981)



MDA character set (1981)

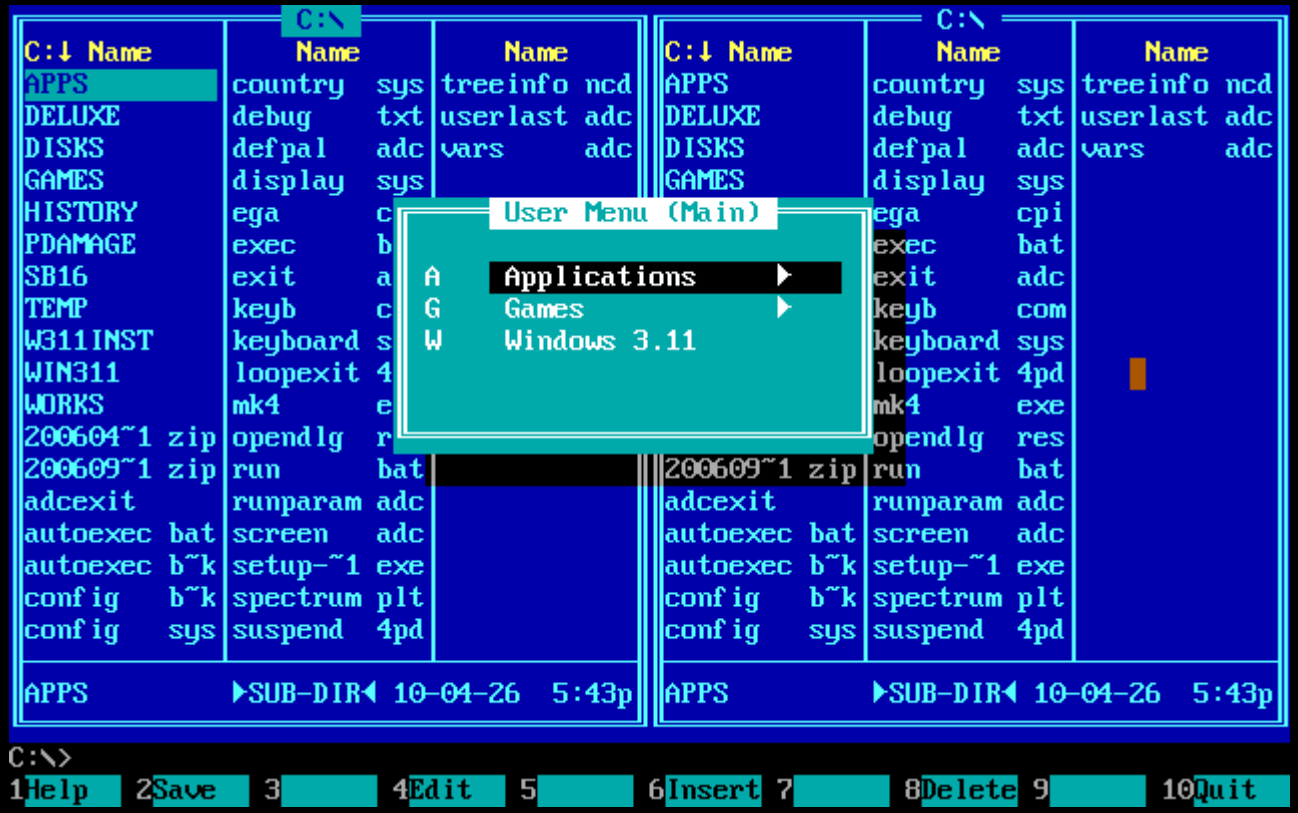


PETSCII character set (1977)
in Commodore 64



ATASCII character set (1979)

Semigraphics



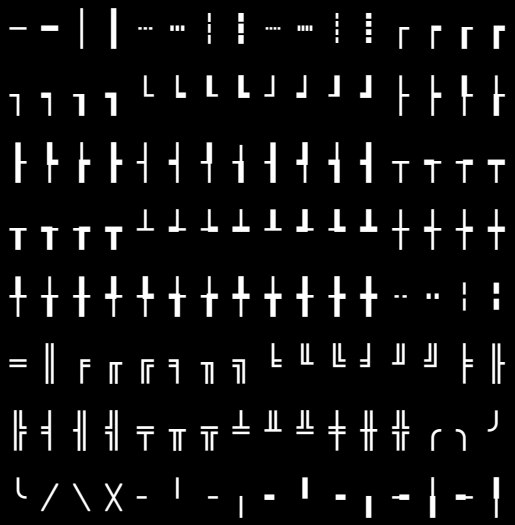
Norton Commander 5.5 (1995)

Unicode - subsets

U+0000..U+007F - ASCII (Basic Latin) Aa Bb Cc Dd … [/]~
U+0080..U+00FF - Latin-1 Supplement ÁáåāőűýñÑ …
U+0100..U+01FF - Latin Extended-A/-B ĀāĂăĄąĆćĈĉĊċ …
…
U+0370..U+03FF - Greek and Coptic Αα Ββ Γγ Δδ … Ωω
U+0400..U+04FF - Cyrillic Аа Бб Вв Гг Дд … Яя
U+0530..U+058F - Armenian ԱբԳդԵզԷըԹժԻ …
U+0590..U+05FF - Hebrew אבגדהוזחטיכ
…
U+AC00..U+D7AF - Hangul Syllables 가각갂갃간갅갆갇갈
(over 11000 chars)
…
U+4E00..U+9FFF - CJK Unified Ideographs (almost 21000 chars!)

Unicode - interesting subsets

U+2500..U+257F - box drawing characters

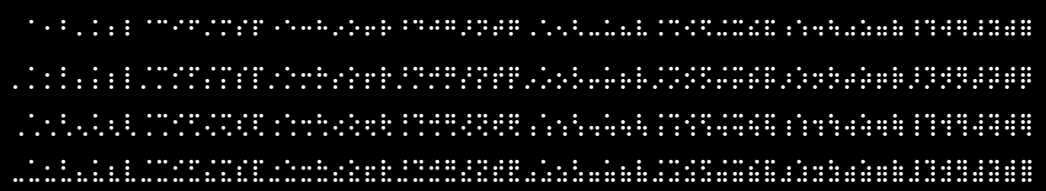


Unicode - interesting subsets

U+2580..U+259F - block elements



U+2800..U+28FF - Braille Patterns (version >= 3.0)



Unicode - interesting subsets

U+1FB00..U+1FBFA - Symbols for Legacy Computing (version >= 13.0, since 2020)

```
>>> for hi in range(0x1fb00, 0x1fbfa, 0x20): print(" ".join([chr(c) for c in range(hi+0, hi+32)]))
```



Teletext semigraphics
(remember Telegazeta?)

Unicode - interesting subsets

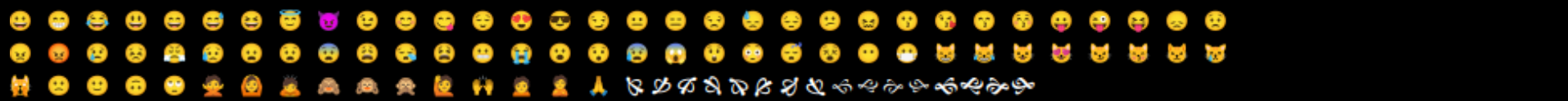
U+2700..U+27bf - Basic emojis (versions 1.0.0 to 7.0, latest 2014)

```
>>> for hi in range(0x2700, 0x27bf, 0x20): print(" ".join([chr(c) for c in range(hi+0, hi+32)]))
...
```



U+1F600..U+1F64F - Basic emojis (version >= 6.0, since 2010)

```
>>> for hi in range(0x1f600, 0x1f650, 0x20): print(" ".join([chr(c) for c in range(hi+0, hi+32)]))
...
```



PART III: STATE OF THE ART

ANSI X3.64 - More colors (xterm-256)

ESC[38;5;*nnn*m
ESC[48;5;*nnn*m

0..7 - standard colors (as for 30..37, 40..47)
8..15 - bright colors (as for 90..97, 100..107)
16..231 - 6x6x6 colors
color(r,g,b) = 36*r + 6*g + b + 16
0 <= r,g,b <= 5
232..255 - 24 shades of gray

„256 colors mode" - actually less, probably 234
e. g. #0 == #16 == #232 (black)
#15 == #231 == #255 (white)

ANSI X3.64 - Even more colors (Tc)

ESC[38;2;*red*;*green*;*blue*m
ESC[48;2;*red*;*green*;*blue*m

red, green and blue: 0..255

„True Color mode” - all 16 777 216 combinations available

ANSI codes and terminal emulators

Text mode	16/8 color mode :((but understands everything)
Konsole, Yakuake	full support
Alacritty	full support, except blinking
PuTTY	full support
CoolRetroTerm	full support (!)
Xterm	full support (!)

ANSI codes and extra characters

Box drawing characters, blocks 2x2:	Unicode 1.0, all systems
Braille 2x4:	Unicode 3.0, all systems
Blocks 2x3, blocks for legacy systems:	Unicode 13.0, not available on Windows (all versions), no problem on Ubuntu

Appropriate font required

TrueColor in tmux

Full support, but tmux tries to be clever…

`\x1b[38;2;0;0;255m` → `\x1b[38;5;21m`

Solution: add to tmux.conf:

```
set -ga terminal-overrides ",original_TERM*:RGB"
set -ga terminal-overrides ",alacritty*:RGB"
```

Useful tools

banner, figlet - generation of text banners in ASCII Art

chafa, jp2a - convert images to text

DurDraw - drawing tool
https://github.com/durdraw/durdraw

PetDraw - drawing tool for Commodore PET, C64, C64 + BI-80, Plus/4
www.the8bitguy.com/download-davids-software/

XTDraw - my own tool inspired by DurDraw and PetDraw
https://github.com/artur-twardowski/xtdraw

lolcat - rainbow colors of any output

\0

(questions?)